

Some Thoughts on Research

SCIE4000, 17 Sep 2025

Nan Ye

School of Mathematics and Physics
The University of Queensland

::maths & industry

::my research

::how to do research

Building a platform for mathematical optimization in the real world

24 April 2024 3:00pm–4:00pm 

Speaker: Andy Ferris
Affiliation: Elara AI

Why is it so hard to get computers to count things? Examples from a Career in Applied Mathematics

10 April 2024 3:00pm–4:00pm 

Speaker: Paul Bellette
Affiliation: Plotlogic

You did what? A walk through the application of Maths and Physics at the Australian Signals Directorate

20 March 2024 3:00pm–4:00pm 

Speaker: Kyla Quinn
Affiliation: Australian Signals Directorate

Quantitative Research in Hedge Funds

10 April 2025 3:00pm–4:30pm 

Speaker: Chao Zhou
Affiliation: Qube RT

Maths for fun and profit (but mostly profit)

16 October 2024 3:00pm–4:00pm 

Speakers: Yannik Rist and Robin Pearce
Affiliation: Biarri Optimization

Machine learning guided scientific experiments

6 September 2023 3:00pm–4:00pm

Speaker: Cheng Soon Ong
Affiliation: CSIRO

Industry Seminar

Life and Times of a Data Scientist

22 May 2025 3:00pm–4:30pm 

Speaker: Virginia Wheway
Affiliation: HCF

AI for improving productivity in work, learning and research

13 March 2024 3:00pm–4:00pm 

Speaker: Brent Richards
Affiliation: Gold Coast HHS/QLD AI Hub

How can we make better decisions with mathematics?

24 September 2019 3:00pm–4:00pm

Speaker: Evan Shellshear
Affiliation: Biarri Optimisation





THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

School of Mathematics & Physics

Industry Seminar Series

Talks by industry speakers

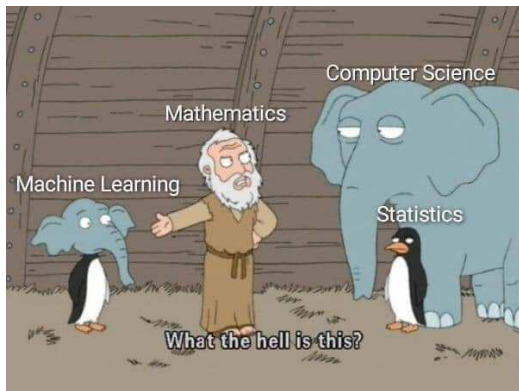
Forge Connections & Your Career

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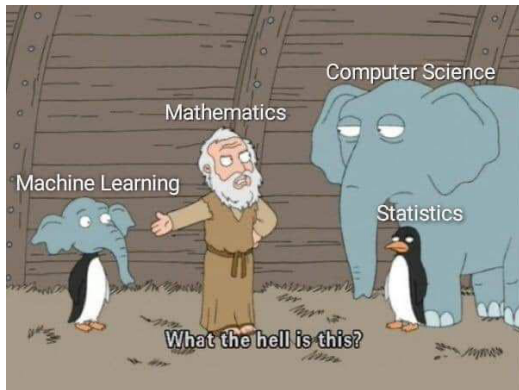


Summer Industry Experience program

big picture #1



big picture #1



Looks like the person who made this wanted to say machine learning is just taking something from computer science and something from statistics, and ended up being not recognised by either community and scorned by mathematics. . .

This might be the view held by quite a few people in the past, but perhaps not any more - the picture would be something like machine learning is a rocket, pulling mathematics, computer science, statistics, biology, physics, ..., up to the sky :D

email to a colleague

big picture #2



an attempt by AI to draw my picture

big picture #3 (my work)



autonomous driving



routing behavior analysis

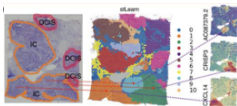


fishery stock assessment

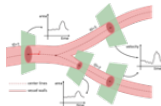


MRI reconstruction

theoretically grounded practical
algorithms
for
learning & decision-making

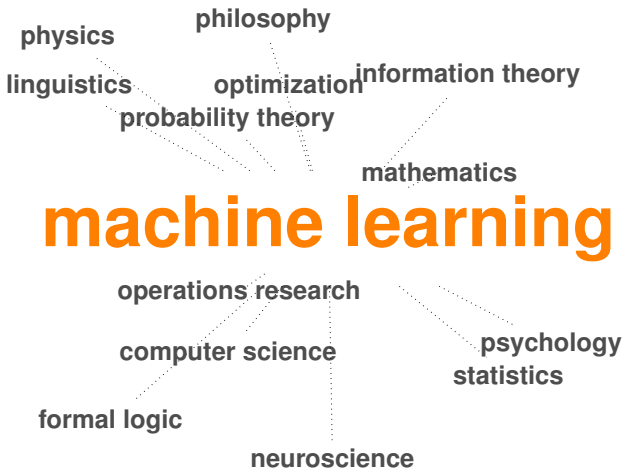


spatial transcriptomics



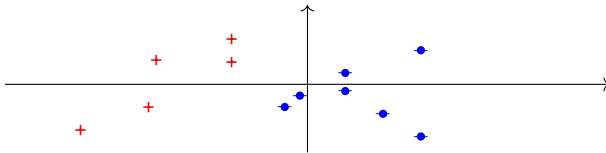
haemodynamics modeling

big picture #4 (synergy)



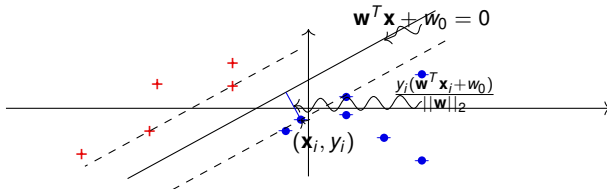
many ideas inspired by other fields & many applications in other fields

an example



question. what's the best line to separate the red and blue points?

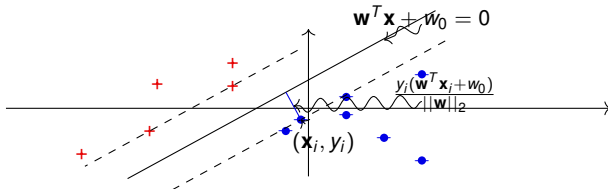
an example



question. what's the best line to separate the red and blue points?

intuition. support vector machines: the one farthest away from all points.

an example



question. what's the best line to separate the red and blue points?

intuition. support vector machines: the one farthest away from all points.

math.

$$\begin{aligned} \min_{\mathbf{w}, w_0} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + w_0) \geq 1, \quad i = 1, \dots, n. \end{aligned}$$

algorithm. how to compute a solution to the optimization problem? $\Rightarrow$ SMO and more...

extensions. what happens if there is no separating line? $\Rightarrow$ slack variables, kernel trick

theory. does this work? $\Rightarrow$ statistical learning theory

practice. applications on real classification datasets

some thoughts

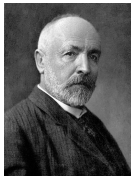
#1 find good questions

they are important

you want to solve them

neither too difficult nor too easy

...



In mathematics the art of proposing a question must be held of higher value than solving it.

Georg Cantor

#2 a failed attempt is a useful finding

a failed attempt eliminates a possible solution

analyzing how things fail can lead to insights on how to make things work



*Once you eliminate the impossible,
whatever remains, no matter how im-
probable, must be the truth.*

Arthur Conan Doyle

#3 balance intuition and rigor

intuition allows us to quickly see a path to a solution, but it may be wrong
rigor is needed for correctness, but often tedious and hides the idea



It is by logic that we prove, but by intuition that we discover.

Henri Poincaré

maths & industry

machine learning research: big pictures, an example

some thoughts

#1 find good questions

#2 a failed attempt is a useful finding

#3 balance intuition and rigor